

# കേരളപ്പിറവി ആശംസകൾ

Kerala Piravi ashamsakal

1 Nov 1956 → 1 Nov 2026 · Kerala turns 70

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# AI · From Hype to BLDing

## your first voice agents

Abhishek Mishra · @stalwartcoder

DevRel & DevEx · Smallest AI · BUILD Day Cochin

TODAY

# Four things, in this order

1. **Build** your first voice agent · ~12 min → laptops out
2. **Look underneath**, with open source · ~12 min
3. **Demo** · call Chechi, the Manglish agent · ~5 min → phone
4. **Your turn** · test it, take the repo home

STEP 0 · GET IN

# Scan, sign up, you’re in



app.smallest.ai

Any email works

Free to start, no card

Slow wifi? One phone hotspot for three of you

BUILD ALONG · FOLLOW MY SCREEN

# Build Kochi Buddy

**1** · Create Agent (green, top right) → Start from Scratch → Single Prompt

**2** · Name it `kochi-buddy` · Call Direction Inbound

**3** · Pick a voice, hit preview · Malayalam (Beta) is in Languages, try it

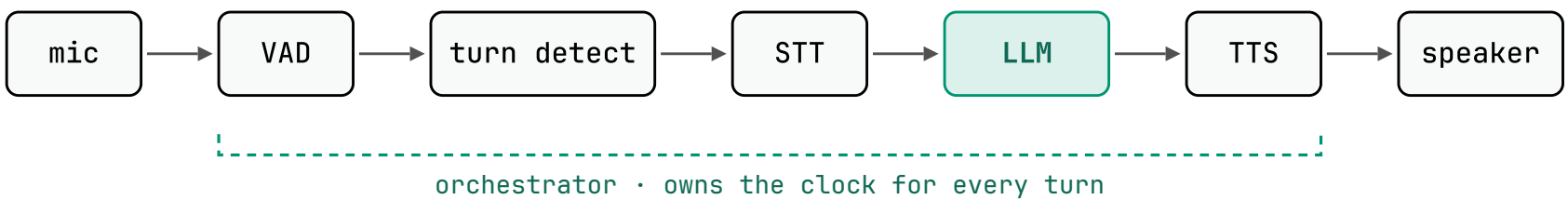
**4** · Paste the prompt from the repo · add a first message · Save

**5** · Test Agent → Web Call → allow the mic → talk

> Try three things. A normal question. A weird one ("book me a flight to Mars"). Then interrupt it mid sentence.

THE BRIDGE

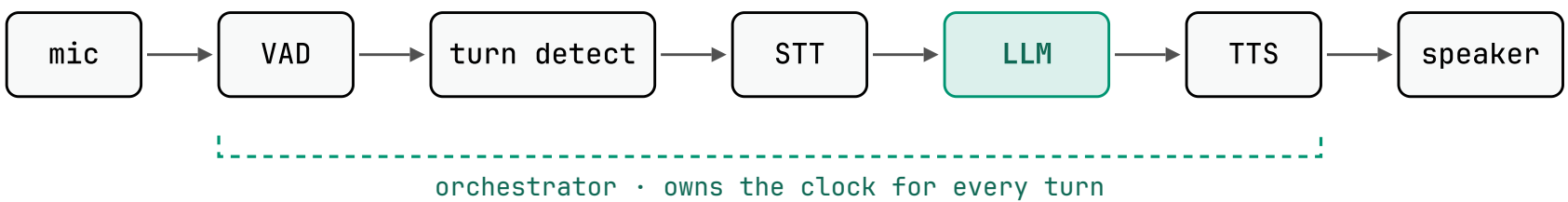
# You just used a platform. Here's what's underneath.



THE MENTAL MODEL EVERYONE STARTS WITH




THE ONE YOU ACTUALLY NEED




**Everything hard about voice AI lives in the arrows, not the boxes.**

**~200ms**

average gap between human turns, 10 languages

Stivers et al., PNAS 2009

**1,500ms**

"an important target to aim for", voice to voice

Voice AI & Voice Agents primer

**?ms**

what your agent actually does

you'll feel it on the phone demo

# Everything must stream

✗ `record → transcribe → think → synthesize → play`

✓ `stream → stream → stream → stream → stream`

Every component emits partial results. Nothing waits for "done".

mic | VAD | turn | STT | LLM | TTS | spk

STOP 1 · TRANSPORT

# How your voice reaches the code

| | Use for | Why |
|---|---|---|
| **WebRTC** | Browser, mobile | Opus codec, jitter buffer, packet-loss concealment, echo cancellation |
| **WebSocket** | Server to server, telephony bridges | Simple, no media features built in |
| **SIP / PSTN** | Phone calls | How the phone network works. Narrowband audio (8 kHz). |

> Do not build WebRTC yourself. NAT traversal, STUN/TURN, ICE, codec negotiation.

mic | VAD | turn | **STT** | LLM | TTS | spk

STOP 2 · VAD, THEN SPEECH TO TEXT

# Your sentence becomes words, while you speak

**VAD** decides "is someone speaking right now?" It gates everything downstream.

**Streaming STT** emits partial transcripts as audio arrives. Not after you stop.

> Batch-first models (Whisper-style) want the whole utterance. A live conversation can't give them that.

Open source · Silero VAD · Whisper · AI4Bharat IndicConformer

mic VAD turn STT **LLM** TTS spk

STOP 3 · THE LLM, PROMPTED FOR THE EAR

# The reply must be speakable

| Text agent | Voice agent |
|---|---|
| Bullet points, markdown, emoji | Never |
| "₹1,234.50" | "twelve hundred thirty-four rupees and fifty paise" |
| Long, thorough answers | 1 or 2 sentences, then stop |
| "Let me know if you need anything else!" | Once. Not every turn. |

mic VAD turn STT LLM **TTS** spk

STOP 4 · TEXT TO SPEECH, THEN PLAYBACK

# The reply becomes a voice

TTS starts on the **first chunk** of LLM output, not the full reply.

Audio streams back over the same transport and plays while the rest is still being generated.

> The metric that matters is **TTFB**, time to first byte. Not total synthesis time.

Open source · Piper · Kokoro · IndicF5




THE WHOLE AGENT AS CODE · PIPECAT

# One pipeline, interchangeable models

```
stt = SmallestSTTService(api_key=SMALLEST_KEY)                    # Pulse
llm = AnthropicLLMService(settings=AnthropicLLMService.Settings(model="claude-haiku-4-5-20251001"))
tts = SmallestTTSService(api_key=SMALLEST_KEY, settings=SmallestTTSSettings(voice="meher"))

# the swap: one line, pipeline untouched
# tts = SmallestTTSService(api_key=SMALLEST_KEY, settings=SmallestTTSSettings(voice="nolan"))
# llm = OpenAILLMService(...)   tts = CartesiaTTSService(...)   stt = DeepgramSTTService(...)

pipeline = Pipeline([transport.input(), stt, ctx.user(), llm, tts,
                     transport.output(), ctx.assistant()])
```

The orchestrator owns transport, turn-taking, interruption and context. **Pipecat** here. **LiveKit Agents** is the other big one. Both are open source.

HARD PART 1 · KNOWING WHEN THE HUMAN STOPPED TALKING

# "I'd like a large pizza…"

…done? Or

# "I'd like a large pizza… and garlic bread."

Silence timer → semantic endpointing → semantic + acoustic. `Pipecat Smart Turn · LiveKit turn detector`

HARD PART 2 · BARGE-IN, AND THE CONTEXT BUG

# What did the agent actually say?

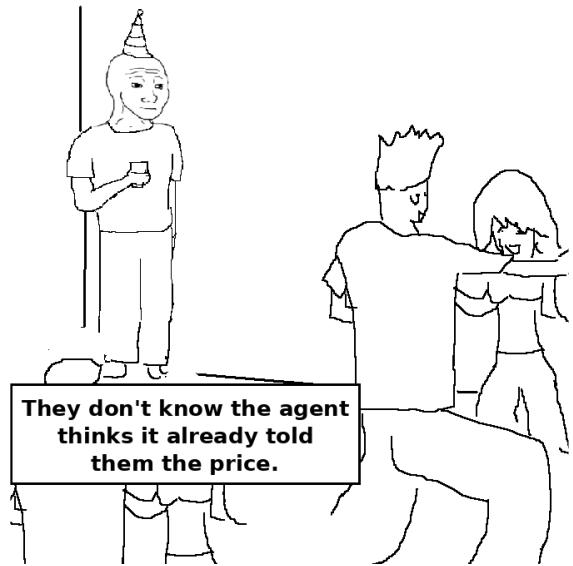
Agent: *"Sure, I can book that for you on Tuesday at 3pm, and I'll send a confirm..."*

User cuts in at **"Tuesday."**

**What does the conversation history say the agent said?**

✕ The full sentence. The agent now believes 3pm is agreed. It is not.

✓ Truncate to what was **actually played out loud**.



THE SHOPPING LIST

# Ten building blocks

**1 · Transport**
WebRTC / WS / SIP. How audio moves.

**2 · VAD + turn detection**
When to listen, when to speak.

**3 · STT**
The ears. Streaming, always.

**4 · LLM + voice prompt**
The brain, prompted for the ear.

**5 · TTS**
The voice. TTFB is the metric.

**6 · Orchestrator**
The conductor. Pipecat or LiveKit Agents.

**7 · Tools / APIs**
The hands. Async, masked.

**8 · Knowledge / RAG**
The memory. Keep retrieval tight.

**9 · Telephony**
Optional. A number and a carrier.

**10 · Evals + observability**
Latency percentiles. Listen to calls.

The framework hands you 1, 2 and 6. You choose 3, 4, 5. You build 7 and 8. You rent 9. You never stop doing 10.

LIVE · CHECHI SPEAKS MANGLISH

# +91 22 6809 3636

I call, you shout the questions. English, Hindi or Malayalam.

It's an AI. Normal call charges. Try it yourself after the session, one call at a time.

TEST IT

# How do you know it still works tomorrow?

```python
def test_brain_replies_in_manglish_and_keeps_it_short():
    reply = chechi.think("ചേച്ചി, ഇന്ന് Kerala Piravi ആണല്ലോ. ഒരു hi പറയാമോ?")
    assert MALAYALAM.search(reply)          # Malayalam in the reply
    assert len(reply) < 400                 # short enough to say out loud

@pytest.mark.live
def test_live_chechi_answers_the_phone():
    result = sipling.probe(uri, transport="tcp", timeout=15)   # a real SIP call
```

offline: `pytest -v test_chechi_call.py`

phone: `pytest -v --live -k live test_chechi_call.py`

sipling · open source · github.com/abhishekmishragithub/sipling

TAKE IT HOME

# Everything from today is in here



1 · Finish your agent at home

2 · Run Chechi from the repo

3 · Post it, tag `@stalwartcoder` `#BUILDDayCochin`

`github.com/abhishekmishragithub/workshops · kochi-build-day-2026`



# Questions

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DevRel & DevEx · Smallest AI

Nanni, Kochi! Kerala Piravi ashamsakal